

prof分析-基于megatron-sft框架

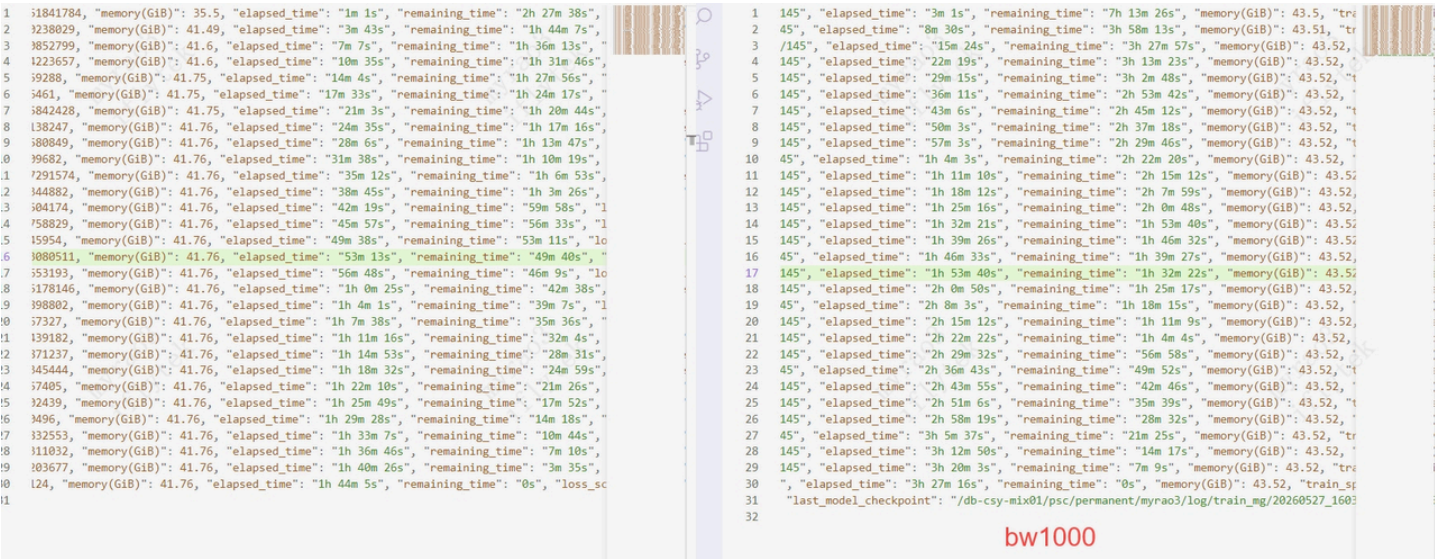
1. 背景

1.1 基础背景

模型：Qwen3-30B-A3B

客户环境： dtk2504.2+cp310+torch2.5

性能表现：



左边是华为910B 右边是BW1000 可以看到 每五次迭代的训练时长约7分钟 友商的约为三分半

1.2 启动脚本

```
source /db-csy-home/psc/myrao3/swift-megatron_v2.bashrc

mkdir -p /tmp/fake_cuda_include
export NVTE_CUDA_INCLUDE_DIR=/tmp/fake_cuda_include

export MODELSCOPE_OFFLINE=1
export SWIFT_USE_MODELSCOPE=0

# sleep 1000
export ENABLE_JIT_FUSER=0 # 0 or 1
# export PYTORCH_HIP_ALLOC_CONF=expandable_segments:True

# MASTER_ADDR=10.121.128.2
# MASTER_PORT=19291
# WORLD_SIZE=8
# RANK=0
export SWIFT_PATCH_CONV3D=False
export ENABLE_JIT_FUSER=0
export THETA_DEVICE="dcu"
export THETA_BACKEND="rcc1"

tmp_cache=/db-csy-mix01/psc/permanent/myrao3/cache

export HF_HOME=$tmp_cache/huggingface/${RANK:-0}
export HF_DATASETS_CACHE=$tmp_cache/huggingface/datasets/${RANK:-0}
export HUGGINGFACE_HUB_CACHE=$tmp_cache/huggingface/hub/${RANK:-0}
export TRANSFORMERS_CACHE=$tmp_cache/huggingface/transformers/${RANK:-0}
export MODELSCOPE_CACHE=$tmp_cache/modelscope/${RANK:-0}
export XDG_CACHE_HOME=$tmp_cache
export TORCH_EXTENSIONS_DIR=$tmp_cache/torch_extensions

mkdir -p "$HF_HOME"
mkdir -p "$HF_DATASETS_CACHE"
mkdir -p "$MODELSCOPE_CACHE"
mkdir -p "$HUGGINGFACE_HUB_CACHE"
mkdir -p "$TRANSFORMERS_CACHE"
mkdir -p "$TORCH_EXTENSIONS_DIR"
```

```
# 禁用 datasets 的文件锁, 防止 NFS 上创建锁文件失败
export HF_HUB_DISABLE_SYMLINKS=1
export HF_HUB_ENABLE_HF_TRANSFER=0

export NCCL_IB_HCA=mlx5_2,mlx5_3,mlx5_4,mlx5_5,mlx5_6,mlx5_7,mlx5_8,mlx5_9
export NCCL_IB_TC=162
export NCCL_IB_GID_INDEX=3
export NCCL_SOCKET_IFNAME=enol
export NCCL_IB_DISABLE=0
export NCCL_P2P_LEVEL=AXB
export NCCL_DEBUG=INFO
# export NCCL_SOCKET_IFNAME=eth0 # 提交单机需打开, 调试机或者提交多机需注释
source /opt/dtk-25.04.2/env.sh
export NCCL_TOPO_FILE=/db-csy-mix01/sft/rccl_tests/temp.xml
export RCCL_TOPO_FILE=/db-csy-mix01/sft/rccl_tests/temp.xml
source /etc/profile.d/modules.sh
module unuse /public/software/modules
module use /opt/software/modules
# module load compiler/dtk/25.04.2

unset TRITON_OVERRIDE_ARCH
unset PYTORCH_ROCM_ARCH
unset AMDGPU_TARGETS

export PYTORCH_ROCM_ARCH=gfx936
export AMDGPU_TARGETS=gfx936
export HSA_OVERRIDE_GFX_VERSION=9.3.6

export TORCH_COMPILE_DISABLE=1
export TORCHDYNAMO_DISABLE=1
export PYTORCH_JIT=0

WORLD_SIZE=$WORLD_SIZE
RANK=$RANK
MASTER_ADDR=$MASTER_ADDR
MASTER_PORT=$MASTER_PORT
```

```
NPROC_PER_NODE=8 \
megatron sft \
  --mcore_model /db-csy-mix01/psc/permanent/myrao3/exp/pretrain_model/mcore_model/Qwen3-30B-A3B-Thinking-2507-mcore \
  --save_safetensors false \
  --load_from_cache_file true \
  --cached_dataset /db-csy-mix01/psc/permanent/myrao3/exp/data/cached_dataset/train_data_10p_v5_round2_1分制/train \
  --split_dataset_ratio 0.001 \
  --tensor_model_parallel_size 4 \
  --pipeline_model_parallel_size 2 \
  --expert_model_parallel_size 4 \
  --split_dataset_ratio 0.001 \
  --sequence_parallel true \
  --moe_grouped_gemm true \
  --moe_shared_expert_overlap true \
  --moe_aux_loss_coeff 1e-6 \
  --micro_batch_size 1 \
  --global_batch_size 64 \
  --recompute_granularity full \
  --recompute_method uniform \
  --recompute_num_layers 1 \
  --packing true \
  --num_train_epochs 3 \
  --finetune true \
  --lr 3e-5 \
  --lr_warmup_fraction 0.05 \
  --min_lr 3e-6 \
  --output_dir $output_dir/megatron_output \
  --save_total_limit 5 \
  --eval_steps 100 \
  --save_steps 100 \
  --max_length 32768 \
  --data_loader_num_workers 8 \
  --dataset_num_proc 8 \
  --no_save_optim true \
  --no_save_rng true \
  --attention_backend flash \
```

2. prof分析

2.1 prof脚本

由于之前没有同事在这个框架下抓取过prof 效仿官方文档上对NPU的抓取：

MEGATRON-SWIFT

快速开始

命令行参数

LoRA训练

多模态模型

Mcore Bridge

GRPO

GKD

Ascend NPU

NPU 性能数据采集

NPU 精度数据采集

CUSTOMIZATION

架构介绍

自定义模型

自定义数据集

BEST PRACTICES

GRPO完整实验流程

多模态GRPO完整实验流程

GRPO代码训练

Qwen3最佳实践

Qwen3-VL最佳实践

NPU 性能数据采集

NPU性能采集通过 `torch_npu.profiler.profile` 接口进行采集，创建`torch_npu.profiler.profile`对象，通过`start`和`stop`接口控制采集性能数据，采集过程需要修改ms-swift源码，修改`swift/megatron/trainers/base.py`文件中的`train`函数，采集示例如下：

```
import torch_npu
...

experimental_config = torch_npu.profiler._ExperimentalConfig(
    profiler_level=torch_npu.profiler.ProfilerLevel.Level1,
    aic_metrics=torch_npu.profiler.AiCMetrics.PipeUtilization,
)

prof = torch_npu.profiler.profile(
    activities=[
        torch_npu.profiler.ProfilerActivity.CPU,
        torch_npu.profiler.ProfilerActivity.NPU
    ],
    schedule=torch_npu.profiler.schedule(wait=0, warmup=0, active=1, repeat=1, skip_first=6),
    on_trace_ready=torch_npu.profiler.tensorboard_trace_handler("./result"),
    profile_memory=False, # 关闭采集内存信息
    with_stack=False, # 关闭采集堆栈信息
    experimental_config=experimental_config)
prof.start()
# ms-swift 逻辑
while state.iteration < args.train_iters:
    ...
    metric, grad_norm, update_successful = train_step(train_data_iterator)
    # 性能数据采集
    prof.step()
    ...
prof.stop()
```

在`base.py`文件中的`train`函数部分插入`prof`代码块 在第五步的时候抓取 先只抓`rank0`（第一张卡）上的性能

代码块

```
1 def train(self, train_dataset, val_dataset):
2     args = self.args
3     config = self.config
4     state = self.state
5     for m in self.wrapped_models:
6         m.train()
7
8     if args.is_multimodal:
9         for m in self.unwrap_models:
10             self._prepare_vit_gradient_checkpointing(m)
11
12     config.grad_scale_func = self.optimizer.scale_loss
13     if isinstance(self.wrapped_models[0], DDP) and
        args.overlap_grad_reduce:
```

```

14         assert config.no_sync_func is None, ('When overlap_grad_reduce is
15         True, config.no_sync_func must be None; '
16         'a custom no_sync_func is not
17         supported when overlapping grad-reduce')
18         config.no_sync_func = [model_chunk.no_sync for model_chunk in
19         self.wrapped_models]
20         if len(self.wrapped_models) == 1:
21             config.no_sync_func = config.no_sync_func[0]
22         if args.align_grad_reduce:
23             config.grad_sync_func = [model_chunk.start_grad_sync for
24             model_chunk in self.wrapped_models]
25             if len(self.wrapped_models) == 1:
26                 config.grad_sync_func = config.grad_sync_func[0]
27         if args.overlap_param_gather and args.align_param_gather:
28             config.param_sync_func = [model_chunk.start_param_sync for
29             model_chunk in self.wrapped_models]
30             if len(self.wrapped_models) == 1:
31                 config.param_sync_func = config.param_sync_func[0]
32         config.finalize_model_grads_func = finalize_model_grads
33         start_iteration = state.iteration
34         pre_hook_enabled = False
35         # Disable forward pre-hook to start training to ensure that errors in
36         # checkpoint loading
37         # or random initialization don't propagate to all ranks in first all-
38         # gather (which is a
39         # no-op if things work correctly).
40         if should_disable_forward_pre_hook(args):
41             disable_forward_pre_hook(self.wrapped_models, param_sync=False)
42             # Also remove param_sync_func temporarily so that sync calls made
43             in
44             # `forward_backward_func` are no-ops.
45             param_sync_func = config.param_sync_func
46             config.param_sync_func = None
47             pre_hook_enabled = False
48
49         self.call_event('on_train_begin')
50         train_metrics = {}
51         if args.virtual_pipeline_model_parallel_size is not None:
52             train_data_iterator, val_data_iterator = [], []
53             for _ in range(args.virtual_pipeline_model_parallel_size):
54                 train_it, val_it = self._prepare_data_iterator(train_dataset,
55                 val_dataset)
56                 train_data_iterator.append(train_it)
57                 val_data_iterator.append(val_it)
58             else:
59                 train_data_iterator, val_data_iterator =
60                 self._prepare_data_iterator(train_dataset, val_dataset)

```



```

51
52     prof_step=10
53     prof_done = False
54     import torch.distributed as dist
55     while state.iteration < args.train_iters:
56         self.call_event('on_step_begin')
57         maybe_finalize_async_save(args, blocking=False)
58
59         if state.iteration == prof_step and not prof_done:
60             rank = dist.get_rank() if dist.is_initialized() else 0
61             if rank == 0:
62                 print(f"\n{'='*80}")
63                 print(f"Starting performance profiling at step {prof_step}
64 (Rank {rank})")
65                 print(f"{'='*80}\n")
66                 with torch.profiler.profile(
67                     activities=[
68                         torch.profiler.ProfilerActivity.CPU,
69                         torch.profiler.ProfilerActivity.CUDA, # DCU使用CUDA
70                     ],
71                     record_shapes=True,
72                     profile_memory=False,
73                     with_stack=True,
74                 ) as prof:
75                     metrics, grad_norm, update_successful =
76 self.train_step(train_data_iterator)
77                     if rank == 0:
78                         print("\n" + "=" * 80)
79                         print(f"Performance profiling at step {prof_step}")
80                         print("=" * 80)
81                         print(prof.key_averages(group_by_input_shape=True).table(
82                             sort_by="self_cuda_time_total",
83                             row_limit=30
84                         ))
85                         prof.export_chrome_trace('./megatron_prof.json')
86                         print("Profile saved to ./megatron_prof.json")
87                         print("=" * 80 + "\n")
88                         prof_done = True
89                         break
90
91         else:
92             metrics, grad_norm, update_successful =
93 self.train_step(train_data_iterator) #除了第十步采集性能 别的都在正常训练
94             if state.iteration == start_iteration:
95                 if update_successful:
96                     # Enable forward pre-hook after training step has
97 successfully run. All subsequent

```

```

94             # forward passes will use the forward pre-hook /
`param_sync_func` in
95             # `forward_backward_func`.
96             if should_disable_forward_pre_hook(args):
97                 enable_forward_pre_hook(self.wrapped_models)
98                 config.param_sync_func = param_sync_func
99                 pre_hook_enabled = True
100         else:
101             start_iteration = state.iteration + 1
102
103         state.iteration += 1
104         self.call_event('on_step_end')
105         self._aggregated_metrics(metrics, train_metrics)
106         train_metrics['grad_norm'] = grad_norm
107         learning_rate = None
108         for param_group in self.optimizer.param_groups:
109             if len(param_group['params']) == 0:
110                 continue
111             learning_rate = param_group['lr']
112         if learning_rate is not None:
113             train_metrics['learning_rate'] = learning_rate
114         if state.should_log:
115             state.should_log = False
116             self.on_log(logs=train_metrics)
117             train_metrics = {}
118
119         eval_metrics = None
120         if state.should_eval:
121             state.should_eval = False
122             if should_disable_forward_pre_hook(args):
123                 disable_forward_pre_hook(self.wrapped_models)
124                 pre_hook_enabled = False
125             eval_metrics = self.evaluate(val_data_iterator)
126             for m in self.wrapped_models:
127                 m.train()
128             if should_disable_forward_pre_hook(args):
129                 enable_forward_pre_hook(self.wrapped_models)
130                 pre_hook_enabled = True
131
132         if state.should_save:
133             self._determine_best_metric(eval_metrics)
134             if should_disable_forward_pre_hook(args):
135                 disable_forward_pre_hook(self.wrapped_models)
136             state.should_save = False
137             self.save_checkpoint()
138             if should_disable_forward_pre_hook(args):
139                 enable_forward_pre_hook(self.wrapped_models)

```

```
140
141         self.call_event('on_train_end')
142         # Close out pre-hooks if using distributed optimizer and overlapped
param gather.
143         if pre_hook_enabled:
144             disable_forward_pre_hook(self.wrapped_models)
145         maybe_finalize_async_save(args, blocking=True, terminate=True)
```

2.2 prof分析及优化

抓取后直接拉表分析

Perfetto
⋮

CURRENT TRACE ↑

megatron_prof.json (2161 MB)

- Timeline
- Overview
- Data Explorer
- Query (SQL)**
- Metrics
- Download

NEW TRACE ↑

- Open trace file
- Record new trace
- Open Android example
- Open Chrome example

SETTINGS ↑

- Settings
- Flags
- Plugins

SUPPORT ↓

Documentation & Bugs

CONVERT TRACE ↓

WSM SW ON v55.3-df2b6e4be

Search or type '>' for commands, ':' for SQL query

<> Query 1 +

▶ Run Query or press ⌘ R

```
1 SELECT name,COUNT(*) AS record_count,SUM(dur) / 1000000.0 AS total_duration_ms, SUM(dur) / COUNT(*) / 1000000.0 AS avg_duration_ms FROM s
```

Returned 183 rows in 269 ms

name	record_count	total_duration_ms ↓	avg_duration_ms
ncclDevKernel_SendRecv_ring_simple_sum_int8_t(ncclDevKernelArgsStorage<4096ul>)	3492	24637.531706	7.055421
ncclDevKernel_AllGather_Ring_Simple_Sum_i8(ncclDevKernelArgsStorage<4096ul>)	1940	10422.446058	5.372394
ncclDevKernel_ReduceScatter_Ring_Simple_Sum_b16(ncclDevKernelArgsStorage<4096ul>)	786	5330.593418	6.781925
Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT32x32x32_MI16x16x1_SN_AFC1_DTL0_GRVWA4_GRVWB4...	10015	2879.183615	0.287487
void at::native::vectorized_elementwise_kernel<4, at::native::CUDAFunctionAdd<c10::BFloat16>, at::...	33036	2757.678857	0.083474
void at::native::(anonymous namespace)::CatArrayBatchedCopy_contig_cat::native::(anonymous na...	2303	1943.648422	0.843963
Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT128x128x32_MI16x16x1_SN_AFC1_DTL0_GRVWA8_GRVW...	4748	1849.928684	0.389622
Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT128x64x32_MI16x16x1_SN_AFC1_DTL0_GRVWA8_GRVWB...	11343	1687.812327	0.148797
void at::native::scatter_gather_elementwise_kernel<256, 4, at::native::cuda_scatter_gather_intern...	762	1473.010105	1.933084
Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT128x128x32_MI16x16x1_SN_AFC1_DTL0_GRVWA8_GRVW...	4932	1358.656462	0.275477
Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT128x64x32_MI16x16x1_SN_AFC1_DTL0_GRVWA8_GRVWB...	8747	1356.79291	0.155115
Custom_Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT256x256x16_MI16x16x1_SN_GSU1_K1_LRWV2_M...	4967	1332.376194	0.268245
void at::native::elementwise_kernel<128, 4, at::native::gpu_kernel_impl_nocast_cat::native::BinaryFu...	3837	1312.985475	0.34219
Custom_Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT256x256x16_MI16x16x1_SN_DTL1_GSU1_K1_MI...	11347	1267.914365	0.11174
Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT32x64x32_MI16x16x1_SN_AFC1_DTL0_GRVWA4_GRVWB8...	3151	1248.689629	0.396283
Cijk_Alik_Blijk_BBS_BH_Bias_AS_SAV_MT256x256x32_MI16x16x1_SN_AFC1_DTL0_GRVWA8_GRVW...	3797	1165.284863	0.306896
ncclDevKernel_AllGather_Ring_LL_Sum_i8(ncclDevKernelArgsStorage<4096ul>)	768	1116.447066	1.453707
void at::native::elementwise_kernel<128, 4, at::native::gpu_kernel_impl_cat::native::BinaryFunction<fl...	2299	1076.841358	0.468395
void at::native::indexFuncLargeIndex<c10::BFloat16, long, unsigned int, 2, 2, -2, true, at::native::(a...	378	1068.377539	2.826395

可以看到 耗时排名前几的kernel都是通信 但是通信测试没有问题

从左往右 从上往下分别是 All reduce /all to all / all gather与reduce scatter

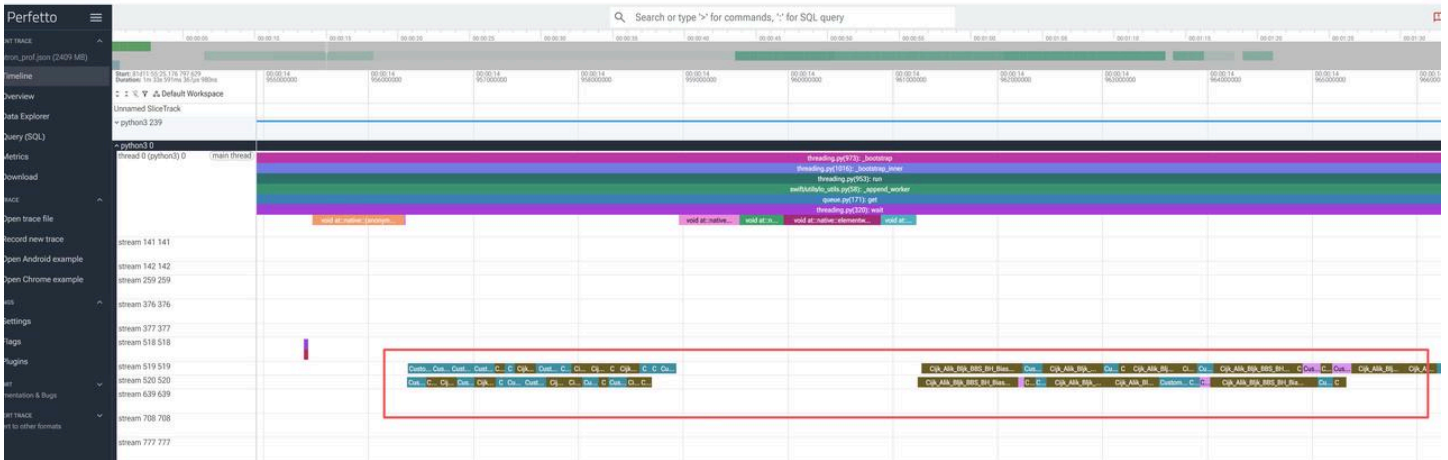
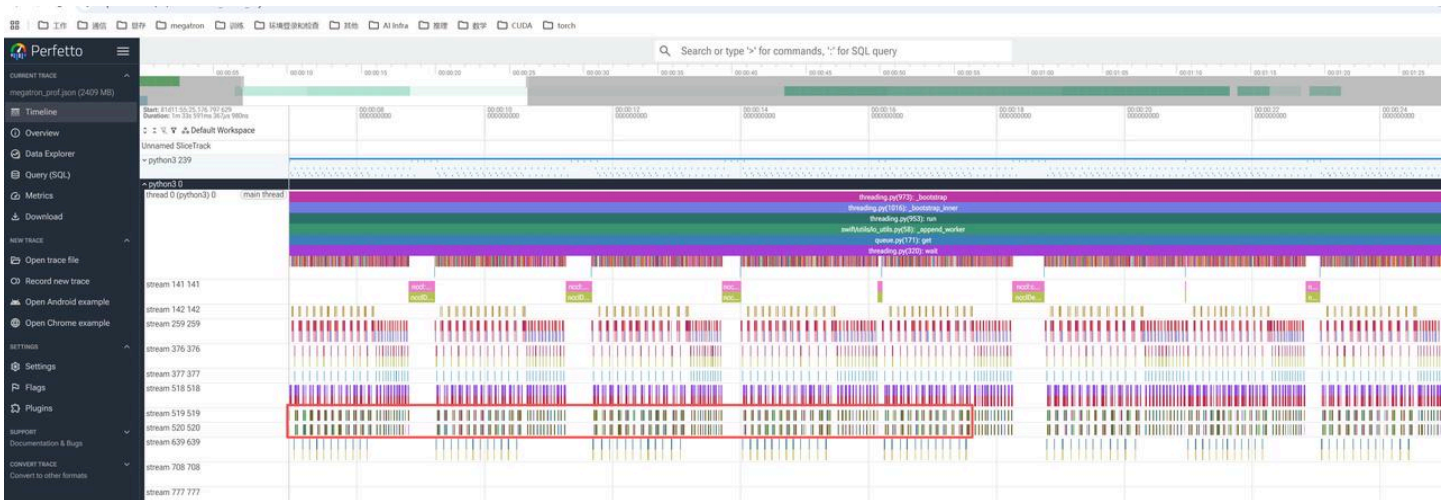
[illegible]

#	size	count	type	redop	root	time	out-of-place	time	in-place	#wrong
#	(B)	(elements)				(us)	alg/bw	(us)	alg/bw	
0	0	0	float	none	-1	0.16	0.00	0.00	0.15	0.00
0	0	0	float	none	-1	0.15	0.00	0.00	0.15	0.00
0	0	0	float	none	-1	0.15	0.00	0.00	0.15	0.00
0	0	0	float	none	-1	0.15	0.00	0.00	0.15	0.00
0	0	0	float	none	-1	0.15	0.00	0.00	0.15	0.00
256	4	4	float	none	-1	59.00	0.00	0.00	58.73	0.00
512	8	8	float	none	-1	58.63	0.01	0.01	58.57	0.01
1024	16	16	float	none	-1	59.05	0.02	0.02	58.87	0.02
2048	32	32	float	none	-1	58.35	0.04	0.03	58.31	0.04
4096	64	64	float	none	-1	58.58	0.07	0.07	58.41	0.07
8192	128	128	float	none	-1	59.38	0.14	0.13	58.41	0.14
16384	256	256	float	none	-1	59.36	0.28	0.26	59.36	0.28
32768	512	512	float	none	-1	64.56	0.51	0.48	64.33	0.51
65536	1024	1024	float	none	-1	69.47	0.94	0.88	69.19	0.95
131072	2048	2048	float	none	-1	70.42	1.86	1.75	69.33	1.89
262144	4096	4096	float	none	-1	75.97	3.45	3.23	75.61	3.47
524288	8192	8192	float	none	-1	115.0	4.56	4.27	110.2	4.76
1048576	16384	16384	float	none	-1	203.7	9.16	8.83	198.9	9.28
2097152	32768	32768	float	none	-1	238.8	17.38	16.23	238.8	17.38
4194304	65536	65536	float	none	-1	243.0	17.23	16.15	243.5	17.23
8388608	131072	131072	float	none	-1	261.1	32.05	30.16	261.1	32.05
16777216	262144	262144	float	none	-1	281.3	59.85	55.92	333.7	50.28
33554432	524288	524288	float	none	-1	369.7	90.77	85.10	368.5	91.05
67108864	1048576	1048576	float	none	-1	593.1	113.52	106.42	591.2	113.51
134217728	2097152	2097152	float	none	-1	1053.1	127.46	119.49	1084.3	127.46
268435456	4194304	4194304	float	none	-1	2041.8	131.47	123.25	2002.0	134.88
536870912	8388608	8388608	float	none	-1	3987.7	134.63	125.22	3948.6	135.96
1073741824	16777216	16777216	float	none	-1	7882.4	136.22	127.71	7843.3	136.90
2147483648	33554432	33554432	float	none	-1	15624	137.19	128.61	15664	137.10

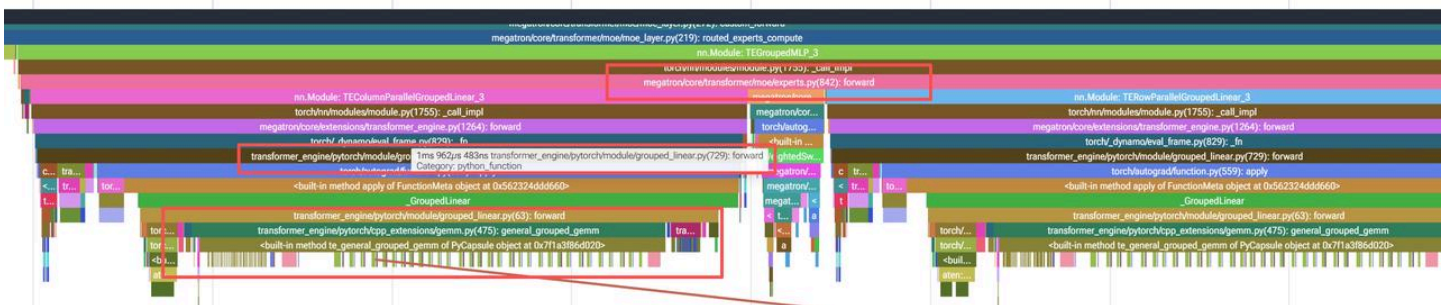
#	(B)	(elements)	sum	(us)	(GB/s)	(GB/s)	sum	(us)	(GB/s)	(GB/s)
0	0	0	float	-1	1.56	0.00	0.00	0	0.14	0.00
0	0	0	float	-1	0.14	0.00	0.00	0	0.14	0.00
0	0	0	float	-1	0.14	0.00	0.00	0	0.14	0.00
0	0	0	float	-1	0.14	0.00	0.00	0	0.14	0.00
0	0	0	float	-1	0.14	0.00	0.00	0	0.14	0.00
256	4	4	float	-1	62.04	0.00	0.00	60.98	0.00	0.00
512	8	8	float	-1	61.63	0.01	0.01	61.55	0.01	0.01
1024	16	16	float	-1	63.05	0.02	0.02	62.86	0.02	0.02
2048	32	32	float	-1	62.73	0.03	0.03	62.64	0.03	0.03
4096	64	64	float	-1	62.88	0.07	0.06	63.14	0.06	0.06
8192	128	128	float	-1	63.53	0.13	0.12	63.23	0.13	0.12
16384	256	256	float	-1	63.77	0.26	0.24	63.85	0.26	0.24
32768	512	512	float	-1	64.97	0.50	0.47	65.08	0.50	0.47
65536	1024	1024	float	-1	69.67	0.94	0.88	69.41	0.94	0.89
131072	2048	2048	float	-1	70.27	1.87	1.75	69.54	1.88	1.77
262144	4096	4096	float	-1	75.97	3.45	3.24	75.68	3.45	3.25
524288	8192	8192	float	-1	109.2	4.80	4.50	109.7	4.78	4.48
1048576	16384	16384	float	-1	202.1	5.19	4.86	198.6	5.28	4.95
2097152	32768	32768	float	-1	237.0	8.45	8.20	236.9	8.45	8.20
4194304	65536	65536	float	-1	241.1	17.40	16.31	240.7	17.42	16.33
8388608	131072	131072	float	-1	260.5	32.20	30.19	260.2	32.24	30.22
16777216	262144	262144	float	-1	280.3	59.85	55.11	280.5	59.81	55.07
33554432	524288	524288	float	-1	371.4	90.36	84.71	371.1	90.42	84.77
67108864	1048576	1048576	float	-1	590.2	113.70	106.69	590.3	113.68	106.58
134217728	2097152	2097152	float	-1	1045.9	128.33	120.30	1046.4	128.27	120.25
268435456	4194304	4194304	float	-1	1997.1	134.42	126.61	1994.5	134.59	126.18
536870912	8388608	8388608	float	-1	3979.7	134.90	126.47	3933.0	136.50	127.97
1073741824	16777216	16777216	float	-1	7890.6	137.61	129.01	7892.1	137.52	129.02
2147483648	33554432	33554432	float	-1	15527	136.30	129.66	15520	136.37	129.72

Errors with asterisks indicate errors that have exceeded the maximum threshold.

分析trace后发现 海光trace文件中出现大量小gemm算子



以上矩阵乘对应栈：



添加环境变量export NVTE_USE_HIPBLASLT_GROUPEDGEMM=1后性能翻倍

```
log > train_mg_dtk2604 > 20260622_170756 > v0-20260622-170834 > {} logging.jsonl
```

```
1  "iteration": "1/145", "elapsed_time": "1m 49s", "remaining_time": "4h 20m 27s", "memory(GiB)"
2  "iteration": "2/145", "elapsed_time": "3m 9s", "remaining_time": "3h 45m 39s", "memory(GiB)"
3  "iteration": "3/145", "elapsed_time": "4m 30s", "remaining_time": "3h 33m 12s", "memory(GiB)"
4  "iteration": "4/145", "elapsed_time": "5m 51s", "remaining_time": "3h 25m 68s", "memory(GiB)"
5  "iteration": "5/145", "elapsed_time": "7m 11s", "remaining_time": "3h 21m 8s", "memory(GiB)"
6  "point": null, "best_model_checkpoint": null, "best_metric": null}
7
```

```
log > train_mg_dtk2604 > 20260623_094552 > v0-20260623-094627 > {} logging.jsonl
```

```
1  i3, "iteration": "1/145", "elapsed_time": "1m 49s", "remaining_time": "4h 22m 21s", "memory(GiB)"
2  i91, "iteration": "2/145", "elapsed_time": "2m 33s", "remaining_time": "3h 2m 38s", "memory(GiB)"
3  i93, "iteration": "3/145", "elapsed_time": "3m 17s", "remaining_time": "2h 35m 47s", "memory(GiB)"
4  i381, "iteration": "4/145", "elapsed_time": "4m 2s", "remaining_time": "2h 22m 3s", "memory(GiB)"
5  i725, "iteration": "5/145", "elapsed_time": "4m 46s", "remaining_time": "2h 13m 16s", "memory(GiB)"
6  "checkpoint": null, "best_model_checkpoint": null, "best_metric": null}
7
```